

BRUCE MBUDI

AI/ML Engineer · MLOps & SRE · Agentic Systems
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PROFESSIONAL SUMMARY

AI/ML Engineer and MLOps Specialist with 7+ years of production engineering experience. Builds and deploys end-to-end ML systems: from model training and RAG pipelines to agentic LLM workflows and real-time inference observability. Strong differentiator in MLOps — bringing SRE-grade reliability (Prometheus, Grafana, Loki, Kubernetes) to ML systems that most ML-only engineers cannot operate or monitor in production. Experienced shipping AI into real business constraints, including resource-limited environments, strict data privacy requirements, and high-volume transaction processing.

TECHNICAL SKILLS

AI / ML	LLMs (Claude API, OpenAI, Ollama), LangGraph, LangChain, RAG pipelines, agentic workflow design, fine-tuning, vector databases, Langfuse, MCP (Model Context Protocol), structured output, HITL systems
MLOps	Prometheus (custom exporters for LLM inference), Grafana, Loki, Alertmanager, OpenTelemetry, SLO/SLA monitoring, Docker, Kubernetes (k3s), CI/CD pipelines, Azure Container Registry
Languages	Python, Go, JavaScript/TypeScript, SQL, Bash
Frameworks	FastAPI, Django, Next.js, Node.js, React
Databases	PostgreSQL, MySQL, Redis, Microsoft SQL Server, vector databases
Integrations	Dynamics 365 F&O, Zoho CRM, n8n, Webhook architectures, OAuth2/Azure Entra ID

EXPERIENCE

Senior AI/ML Engineer & Site Reliability Engineer | **TSAVO Real Estate** | Nairobi, Kenya *May 2024 – Mar 2026*

- Architected and deployed Linzi AI — an agentic SRE platform using LangGraph state machines, PromQL/LogQL, and local LLMs (phi3, qwen3, deepseek-r1) to autonomously detect infrastructure anomalies, triage incidents, and surface security threats; designed with confidence-based fallback and human-in-the-loop escalation for high-severity events.
- Built a custom Python Prometheus exporter for Ollama LLM inference metrics across 4 locally deployed models, enabling real-time monitoring of token throughput, latency, and model availability — the first LLM observability layer in the organisation.
- Engineered an M-PESA automation pipeline (n8n + Telegram + local LLM) processing 30,000+ monthly transactions with peaks of 8,000 in a single weekend, eliminating manual payment entry entirely; integrated structured LLM output for transaction parsing and anomaly flagging.
- Designed and owned production observability architecture across 12+ services and 6 integrations: Grafana/Prometheus/Loki/Alertmanager with 25–30 severity-routed alert rules, SLO dashboards, SSH brute-force detection via Loki pipelines, and automated recovery scripts.
- Architected enterprise integration middleware connecting Dynamics 365 F&O, Zoho CRM/Desk, 3CX telephony, and Jibble via event-driven webhooks and OAuth2; built Claude Code + Anthropic API agentic workflows for document processing and intelligent automation.
- Led ERP data migration (Business Central → Dynamics 365 F&O): custom Python ETL pipelines and validation frameworks ensuring integrity across 17,000+ records.

AI/ML Engineer | **Nurika Health** | Lagos, Nigeria (Remote)

Nov 2023 – Feb 2024

- Designed adaptive goal-setting algorithm analyzing historical step data to generate personalized daily targets using ML techniques, improving user engagement metrics.
- Developed referral fraud detection models using statistical analysis and ML to identify and flag fake user networks, reducing fraudulent accounts.
- Deployed containerized ML inference services (Docker + Azure Container Registry), maintaining high availability for real-time prediction endpoints.

Data Acquisition Engineer | [Dtechtive.com](#) | London, UK (Remote) *Nov 2022 – May 2024*

- Scaled enterprise data acquisition pipeline (Python, BeautifulSoup, Selenium) growing dataset library from 2,000 to 17,000+ indexed datasets — a 750% increase.
- Engineered async scraping infrastructure with rate limiting, proxy rotation, and validation frameworks, increasing extraction efficiency by 85% with full ToS compliance.

Full Stack Developer | [Dtechtive.com](#) | London, UK (Remote) *Nov 2021 – Nov 2022*

- Led product design and technical architecture for a locust detection mobile app targeting East African agricultural communities (Next.js + React Native, v1.0 prototype delivered).

Full Stack Developer (Freelance) | [Upwork](#) | Remote *Jan 2019 – Dec 2021*

- Delivered responsive websites and custom WordPress solutions across multiple industries, achieving measurable conversion improvements for clients.

KEY PROJECTS

Ulinzi SRE Agent — *Agentic LLM system for autonomous infrastructure incident response*
github.com/bruceowenga/ulinzi-sre-agent

- LangGraph multi-agent state machine with confidence-based LLM routing, structured output validation, and human-in-the-loop escalation for P1/P2 incidents.
- Async Prometheus polling across 4 independent loops; integrates Loki, Grafana, and ntfy for real-time alerting; 39 tests; validated against live homelab stack with captured incident logs.
- Foundation for Linzi AI — commercial SMB monitoring platform targeting the Kenyan/African market.

wbts (what broke the server) — *Open-source Go CLI for SRE post-incident root cause analysis*
github.com/bruceowenga/wbts

- Correlates logs from journald, dmesg, Docker, Kubernetes, and auth sources into an interactive TUI.
- v0.3.0 · 27 GitHub stars · 15 releases · AUR packaged · actively maintained.

Homelab Observability Stack (Odin Server) — *Production-grade SRE stack with LLM inference monitoring*
blog.brucembudi.dev

- Grafana/Prometheus/Loki/Alertmanager on Docker/CentOS with 8 exporters, SLO monitoring, and triple alerting channels with severity-based routing.
- Custom Python Prometheus exporter for Ollama LLM inference metrics. Documented as a multi-part blog series with real incident logs.

Igniters Student Sponsorship Portal — *Full-stack Django/Next.js platform replacing Excel workflows*

- Manages 500+ students across 100+ schools with RBAC, automated budget generation, bulk payment scheduling, and financial reporting. Deployed via Docker/Nginx/Gunicorn.

Firefly III MCP Server + n8n Automation — *Personal finance automation with M-PESA parsing*

- Custom MCP server for Firefly III with automated M-PESA transaction parsing into structured financial records via n8n/Telegram integration. (In development)

OPEN SOURCE

- wbts: Go CLI for SRE post-incident analysis · v0.3.0 · 27 GitHub stars · github.com/bruceowenga/wbts
- expo/expo and alexa/ask-cli: Documentation and developer experience contributions.

WRITING & COMMUNITY

Technical blog at blog.brucembudi.dev — in-depth guides on homelab SRE, Grafana observability, AI/LLM deployment, agentic systems, and self-hosting. Covers real production incidents and maintenance logs.

EDUCATION

Kenyatta University · Telecommunications & Information Technology, Computer Systems Networking · 2015–2020

REFERENCES

Alfred Maina · Head of IT, TSAVO Real Estate · alfred@tsavo.ke · +254 712 779 402

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